[METRIC]
A-A-59532/5A
March 28, 2006
SUPERSEDING
A-A-59532/5
January 14, 2000

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

FUSE, INCLOSED LINK, SURFACE MOUNT (SM), 250 VOLTS (V) AC/DC, QUICK ACTING

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring the fuses described herein shall consist of this document and the issue in effect of CID A-A-59532.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This commercial item description (CID) uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

AA59532	/05	-	001
CID number	Applicable CID specification sheet (two digits)		Dash number

SALIENT CHARACTERISTICS.

Interface and physical dimensions. Fuses supplied to this CID shall be as specified herein (see figure 1).

Electrical specifications.

Ampere rating. The ampere rating shall be as specified in table I.

Voltage rating. The voltage rating shall be 250 V ac/dc.

Time current characteristics. The time current characteristics shall be as indicated in table II.

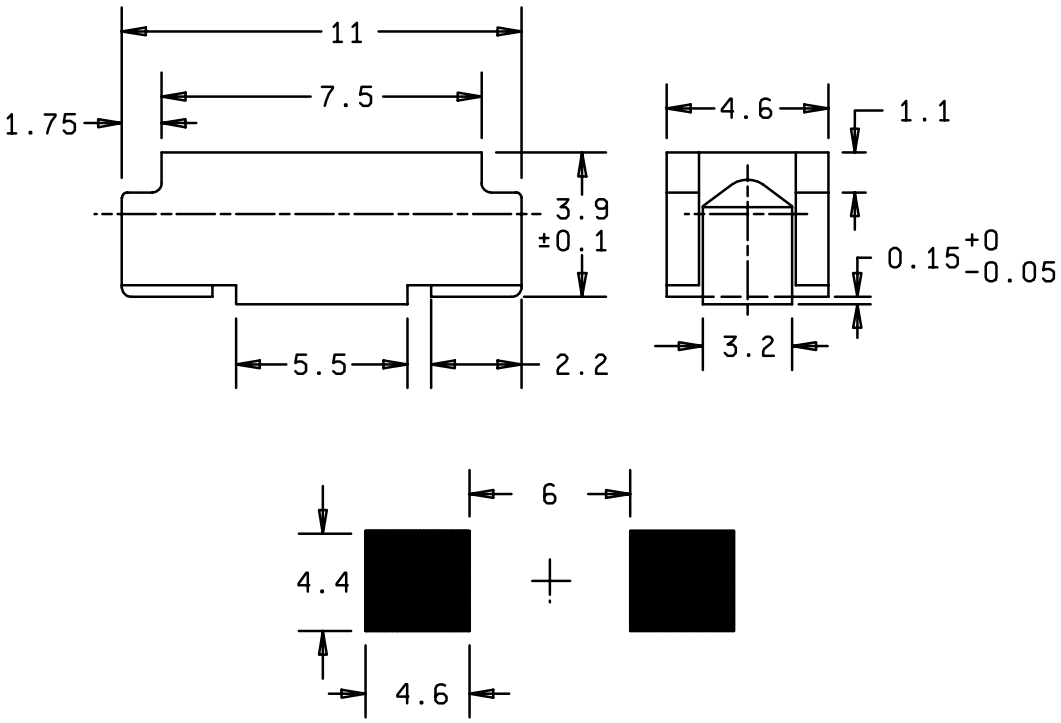
Environmental specifications. Fuses supplied to this CID shall be subject to the following tests and there shall be no electrical or mechanical damage to the fuse.

Maximum storage temperature. The maximum storage temperature shall be 40°C (104°F) at 70 percent relative humidity.

Ambient temperature maximum. The ambient temperature shall be -40°C (-40°F) to +85°C (185°F).

Vibration resistance. Fuses shall meet vibration requirements in accordance with IEC 60068-2-6, Test Fc. (10-2000 Hz, cross-over frequency 60 Hz; < 60 Hz constant Amplitude of 0.75 mm; > 60Hz constant acceleration of 100 m/s^2. Acceleration 196 m/s^2 (20g).

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mm	Inches	mm	Inches
0.1	.004	4.4	.173
0.15	.006	4.6	.181
1.1	.043	5.5	.217
1.75	.069	6	.236
2.2	.087	7.5	.295
3.2	.126	11	.433
3.9	.154		

NOTES:

1. Dimensions are in millimeters.
2. Tolerance is ± 0.13 mm (.005 inch), unless otherwise specified.

FIGURE 1. Configuration and dimensions.

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TABLE I. Electrical characteristics.

AA59532/05-	Ampere rating	Breaking capacity	Voltage drop at rated current. Max. mV	Power dissipation at rated current. typical watts	Fusing I^2t at 4 times rated current	
					Typical A^2s	Maximum A^2s
001	250 mA	100 A/250V ac Power factor (p.f.) / cos φ 1 100A/250Vdc	800	0.110	0.009	0.063
002	315 mA		750	0.125	0.017	0.099
003	400 mA		700	0.19	0.02	0.16
004	500 mA		600	0.19	0.04	0.25
005	630 mA		500	0.23	0.08	0.4
006	800 mA		400	0.33	0.13	0.64
007	1 A		300	0.39	0.23	1
008	1.25 A		300	0.39	0.47	1.53
009	1.6 A		300	0.49	0.84	2.56
010	2 A		300	0.6	1.4	4
011	2.5 A		300	0.67	2.6	6.3
012	3.15 A		300	0.87	4.3	9.9
013	4 A		300	0.95	8.6	16

TABLE II. Opening time characteristics.

Industry standard tested to:	100 percent of rated current	200 percent of rated current	1000 percent of rated current
IEC 60127-4	> 1 hour	≤ 120 seconds	1 ms – 10 ms
UL 248-14/CSA C22.2	> 1 hour	≤ 60 seconds	1 ms – 10 ms

Shock resistance. Fuses shall meet the shock requirements in accordance with IEC 60068-2-27, test Ea (981 m/s^2, 6 ms).

Solderability (reflow and wave soldering). Fuses shall be able to withstand, without electrical or mechanical damage to the fuse, a wave or reflow solder of +235°C (455°F) for 2 seconds maximum in accordance with IEC 60068-2-58 test Td.

Resistance to soldering heat. Fuses shall meet resistance to soldering heat requirements in accordance with IEC 60068-2-58 test Td (10 seconds at 260°C (500 °F)).

Fuse-link temperature rise ≤ 75 K. PC board track widths shall be ≤ 5 mm (0.196 in.) for fuses rated at 5 A or less, ≥ 5 mm (0.196 in.) for fuses rated at 6.3 and 7 A and ≥ 10 mm (.394 in.) for fuses rated at 8 and 10 A.

Permissible continuous operating current at 23°C (73.5 °F). The permissible continuous operating current at 23°C (73.5 °F) shall be 0.8 times rated current.

Physical specifications.

Materials.

Housing material. The fuses shall be temperature resistance plastic with a UL94 VO flammability rating as a minimum.

Terminal plating. Terminals shall be tin-plated brass. Use of pure tin plating is prohibited as a finial finish and as an undercoat. Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent.

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PACKAGING. Packaging shall be as specified in A-A-59532. In addition, these fuses may be supplied individually or in a quantity of 2,000 on a 30.4 mm (1.197 inches) wide tape reel in accordance with IEC 60286-3.

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See classification information for PIN format example.

Source of documents.

Other Publications

Non Government Publications

CANADIAN STANDARDS ASSOCITATION (CSA)

C22.2 - Safety Tests for Low-Voltage Fuses Part 14; Supplemental Fuses, Standard For.

(Copies of this document are available online at http://www.csa.ca/ or from the Canadian Standards Association, 50060 spectrum Way, Mississauga, Ontario L4W5N6 Canada.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60068-2-6 - Environmental Testing – Part 2: Tests – Test Fc: Vibration (Sinusoidal).

IEC 60068-2-27 - Environmental Testing – Part 2: Tests – Test Ea and Guidance: Shock.

IEC 60068-2-58 - Environmental Testing – Part 2-58: Tests – Test Td – Test Methods for Solderability, Resistance to Dissolution of Metallization and to Soldering Heat of Surface Mounting Devices (SMD).

IEC-60127-4 - Miniature Fuses – Part 4: Universal Modular Fuse-Links (UMF) – Through-hole and Surface Mount Types.

IEC 60286-3 - Packaging of Surface Mount Components on Continuous Tape.

(Copies of these documents are available online at http://www.iec.ch/ or from the International Electrotechnical Commission, 3 Rue De Varembe', P.O. Box 131, CH - 1211, Geneve 20, Switzerland.)

UNDERWRITERS LABORATORIES, INCORPORATED (UL)

UL94 - Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances, Standard For.

UL248-14 - Safety Tests for Low-Voltage Fuses Part 14; Supplemental Fuses, Standard For.

(Copies of these documents are available online at http://www.ul.com/ or from the Underwriters Laboratories, Incorporated, 333 Pfingsten Road, Northbrook, IL 60062-2096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

MFR's CAGE	MFR's name and address
61935	Schurter, Incorporated 447 Aviation Boulevard Santa Rosa, CA 95403-1069 Phone number (707) 636-3000 Facsimile number (707) 636-3033 E-mail: mailto:info@schurterinc.com Uniform Resource Locator (URL): http://www.schurterinc.com

Part number (P/N) supersession data. These CID PINs supersede the following MFR's P/N's as shown. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. P/N supersession data.

CID dash number (see table I)	Vendor commercial P/N 1/	
AA59532/05-	MFR's CAGE	MFR's P/N 1/
001	61935	3403.0010.11
002	61935	3403.0011.11
003	61935	3403.0012.11
004	61935	3403.0013.11
005	61935	3403.0014.11
006	61935	3403.0015.11
007	61935	3403.0016.11
008	61935	3403.0017.11
009	61935	3403.0018.11
010	61935	3403.0019.11
011	61935	3403.0020.11
012	61935	3403.0021.11
013	61935	3403.0022.11

1/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID P/N shown. For actual part marking requirements see the marking paragraph of A-A-59532.

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MILITARY INTERESTS:

Custodians:
Navy - EC
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing activity:

DLA-CC

Project 5920-2006-013

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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